Marmara University
Electrical and Electronics Engineering Department
Spring 2023
EE 3012 – Electronics I
Simulation Project - Due April 28, 2023

Operations Using Diodes

The objective of this project is to design and test (by simulation) a diode circuit that produces the difference of two small ac inputs. The ac output of the circuit is given by

$$v_o = A(v_1 - v_2)$$

where v_1 and v_2 are two independent small-amplitude, analog sinusoidal input signals and A is a constant.

Essential background and analysis: Read section 3.3.8 in the textbook.

The required tasks of this project are as follows.

1) Design and test the difference diode circuit (No transistors or ICs). Test the limits of operation of the circuit (in terms of input signal level).
2) Study the range of the constant A. Is it possible to make it greater than 1?
3) Extend to implement the addition operation.
4) Extend to implement the operation $v_o = Av_1 \pm Bv_2$, where A and B are, in general, unequal constants.
5) Find and discuss possible applications of your design.

Instructions

- Use discrete components only.
- Use LTspice only.
- All component types and values should be "real" (i.e., available in market). Use the values such that the output signal is clear and with reasonable amplitude.
- Submissions should be in a formal report format, with an introduction, theoretical analysis, design details, analysis of results and validations, and conclusions.
- IMPORTANT. The project mark will mostly go to the strength of the circuit design and the amount of testing of each design.
- Each student will submit an individual work.
- In case of copied work (circuits, figures, results, etc., even partially), all parties will get nulls.

Show all work and analysis
Only pdf files are acceptable
Late submissions are checked but not graded
Copied submissions represent academic dishonesty and are disqualified